2 **Ans: D**

Ball accelerates constantly down XY upon release due to weight-component down the slope, hence velocity increases uniformly. Once travelling on YZ, ball travels with constant velocity. Inelastic collision at Z means a reduction in KE (i.e. lower speed) and since ball travels in reversed direction now, velocity is now in negative region. When the ball travels up in YX direction before coming to rest momentarily, the ball is again under constant retardation due to weight-component down the slope. Hence the velocity-time gradient at the final segment should be the same as the gradient of the initial motion.